Homework 4: Multiple Linear Regression

ISYE 4031 - Summer 2026 - Problems Only Version

Submission Instructions

Include your full name, Georgia Tech email, and GTID in your submitted work. You may complete the homework using JupyterLite, a local Python or R environment, spreadsheet software, handwritten work where appropriate, or another approved environment. Your submission should clearly answer each question and show enough reasoning, formulas, code output, or supporting work to be graded.

Required Data Files

- delivery_time.csv

Problem 1: Delivery Time for Vending-Machine Routes

A soft-drink distributor records delivery time (`Time`), number of cases stocked (`Cases`), and walking distance (`Distance`) for 25 vending-machine service routes.

Answer parts (a) through (i). Use the multiple regression model with both predictors unless a part asks otherwise.

1(a) Create exploratory plots

Create scatter plots or a scatterplot matrix for `Time`, `Cases`, and `Distance`. Describe the relationships you see.

1(b) Estimate the coefficients using matrix notation

Use $\widehat{\boldsymbol{\beta}} = (X'X)^{-1}X'Y$ with an intercept column. Report the coefficient estimates.

1(c) Fit the multiple regression model and write the equation

Use `statsmodels` to fit `Time ~ Cases + Distance`. Compare the result to your matrix estimate.

1(d) Interpret the coefficients

Interpret the intercept, `Cases` coefficient, and `Distance` coefficient in context. Use conditional language for the slopes.

1(e) Estimate σ^2 and σ

Compute SSE, MSE, and the residual standard error. State the residual degrees of freedom.

1(f) Conduct individual t tests for the slopes

For each slope, test $H_0 : \beta_j = 0$ versus $H_a : \beta_j \neq 0$ at $\alpha = 0.05$. State conclusions in context.

1(g) Construct 95% confidence intervals for both slopes

Report and interpret the intervals.

1(h) Test the overall regression significance

Test $H_0 : \beta_1 = \beta_2 = 0$ at $\alpha = 0.05$. State the p-value or compare the F statistic to the critical value.

1(i) Report R^2 and adjusted R^2

Interpret both values in context.

Problem 2: Projection Matrices

Let $H = X(X'X)^{-1}X'$ be the hat matrix for a full-rank design matrix X. Prove that H and $I - H$ are idempotent. That is, show

$$H^2 = H \quad \text{and} \quad (I - H)^2 = I - H.$$

AI Usage Report

Your submission must include an AI Usage Report following the course AI policy. If you used AI tools, include the tool(s), what you asked them to help with, what output you accepted, changed, or rejected, and how you verified the final work. If prompts or screenshots materially affected your submission, include copied prompt text or screenshots in the report when appropriate. If no AI tools were used, include this statement: “No AI tools were used for this submission. You may include this report at the end of your submission or attach it as a separate file.